

From Obstruction to Design: Fibre Conditions for Monitoring and Indicator Choice

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Abstract

The obstruction calculus says, given a monitoring structure, why viability fails; governance needs the converse — what must be measured for viability to be possible. This paper, the D8 item of the generalization catalogue, develops the design reading on the programme’s working systems. The fibre condition — every observation fibre must contain states whose floor-servable instrument sets intersect — is proved necessary and, in the intersection form, necessary-and-sufficient for one-step admissibility of every singleton belief (verified over all 15 partitions of the four-state target region); the pairwise form of the rule is shown by an exact three-codex instance to be *necessary but not sufficient* off convexity — the one-dimensional Helly protection fails for non-convex instrument codices, with the intersection certificate surviving. Three design rules follow as exact identities: the delay rule (an indicator is acceptable only with period beating worst-case exit: $T_{\text{obs}} < \sigma^*(z_0) = z_0 - 1$ on all 48 audited cells), the bias rule (a biased reading is acceptable only under correction: the trap’s $\frac{21}{100}$ drift bound), and the aggregation rule (an aggregate index is acceptable only if its fibres do not merge policy-incompatible states: the audited belief is aggregate-nonviable, z_1 -only-viable). The coarsest admissible monitoring of the target region costs exactly two cells, by exhaustive enumeration (6/6 check families).

Introduction and relation to the programme

The completed items diagnose; the source catalogue reverses the arrow, and this paper executes the reversal: obstruction as design. The fibre condition, the coarsest-admissible-monitoring problem, and the three indicator rules (delay, bias, aggregation) are stated in the catalogue and here made exact and verified on the audited systems. The successor’s information lattice supplies the semantics of structures; the monitoring companion changes nothing in it.

The fibre condition

For a state x let $\mathcal{R}(x)$ be its floor-servable instrument set. For an observation structure with fibres F :

Proposition 1 (Intersection form: necessary and sufficient). Every singleton belief $\{x\}$ in a target region is one-step admissible under the structure iff every fibre F (within the target) satisfies $\bigcap_{x \in F} \mathcal{R}(x) \neq \emptyset$. The pairwise form ($\mathcal{R}(x) \cap \mathcal{R}(y) \neq \emptyset$ for all $x, y \in F$) is necessary but not sufficient in general.

Proof. Necessity and sufficiency of the intersection form are the read-then-act requirement itself: the cell’s one instrument must serve all of its states. For the pairwise claim: on non-convex codices three pairwise-intersecting sets can have empty total intersection; the exact instance below realizes this, so pairwise compatibility cannot imply the existence of a common instrument. (In one dimension with convex sets, Helly’s theorem would protect the pairwise form — the instance shows exactly where that protection ends.) $\square$

Proposition 2 (Dynamic caveat). Singleton fibre-admissibility is necessary only: a structure admissible for every singleton belief can still obstruct a multi-state belief dynamically. On the audited target the exhaustive search records whether such an instance exists; either outcome is an exact finding.

Proof. Successor cells re-partition, so one-step admissibility of singletons does not compose into multi-step viability; the dynamic statement is governed by the recursion, not the static condition. $\square$

Verified record (6/6 exact check families)

E1. Over all 15 partitions of the target $\{(1, 2), (2, 1), (2, 2), (3, 1)\}$: the intersection form is equivalent to singleton admissibility, instance by instance. The non-convex counter- instance: $\mathcal{R}_1 = \{\frac{2}{10}, \frac{6}{10}\}$, $\mathcal{R}_2 = \{\frac{2}{10}, \frac{65}{100}\}$, $\mathcal{R}_3 = \{\frac{6}{10}, \frac{65}{100}\}$ — pairwise intersections nonempty, total intersection empty.

E2 (an exact absence). The dynamic caveat’s exhaustive search over 15 partitions $\times$ target pairs found *no* instance on this target: the singleton rule happens to be sufficient here; the absence is recorded as the finding (necessity-only remains the general claim).

E3. The coarsest admissible monitoring costs exactly 2 cells (7 of 15 partitions admissible); a minimizing partition isolates the demanding state (1, 2) and merges the rest — enumeration certifies minimality.

E4 (delay rule). On the 48-cell grid with the declared hold class: viability at deadline $T_{\text{obs}} \in \{1, 2, 3\}$ is exactly the identity $z_0 \geq 1 + T_{\text{obs}}$, i.e. $T_{\text{obs}} < \sigma^*(z_0)$ — monitoring frequency must beat worst-case exit.

E5 (bias rule). The trap’s uncorrected reading drives every class law to drift $\geq \frac{21}{100}$ (exit); correction restores — biased indicators are acceptable only under correction.

E6 (aggregation rule). The audited belief $\{(1, 2), (2, 1)\}$ is nonviable under the aggregate index and viable under the z_1 -only index: aggregate indicators are acceptable only if their fibres do not merge policy-incompatible states.

What is not claimed

No sensor-cost theory is developed (the cost function is cell count); no informational-efficiency or mechanism-design claims; the delay rule is stated for the audited hold class, not for general exit-time values beyond the successor’s; the design search is exhaustive over the fifteen-partition target, not over general partition lattices (Bell-number growth is acknowledged as the combinatorial wall). Non-convexity enters only as the recorded counterinstance — no general non-convex design theory is claimed. Real monitoring systems involve noise and delay stochastics, which belong to the stochastic selector’s setting.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `monitoring_design_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.